

Muhammad Owais, *Ph.D.*

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📍 Hachioji Shi Matsugaya Tokyo, Japan



Education

- April 2025 – Present ■ **Tokyo Metropolitan University (TMU), Ph.D.**
Major: Computer Science
Advisor: Prof. Dr. Masaki Aida
- Sep 2021 – Sept 2023 ■ **University of Engineering and Technology (UET), Lahore, M.Sc.**
Major: Computer Science
Advisor: Prof. Dr. Muhammad Shoaib
Thesis title: *A Technique To Enhance Social Media Reliance By Detecting Social Bots.*
- Oct 2017 – Aug 2021 ■ **University of Sargodha, BS.**
Major: Computer Science
Advisor: Prof. Muhammad Bilal
Thesis title: *Planting the Seeds of Sustainability with Plantobook.*

Research Publications

Peer-Reviewed Journal Articles

- Shaikh, J. A., Wang, C., Sima, M. W. U., Arshad, M., **Owais, M.**, Hassan, D. S., et al. (2025). "A deep reinforcement learning-based robust intrusion detection system for securing IoMT healthcare networks." *Frontiers in Medicine*.
- Shaikh, J. A., Wang, C., Muhammad, W. U. S., Arshad, M., & **Owais, M.** (2024). "RCLNet: An Effective Anomaly-Based Intrusion Detection for Securing the IoMT System." *Frontiers in Digital Health*.
- **Owais, M.**, Shoaib, M., & Waseem, M. (2023). "A Holistic Approach for Detecting Socialbots on Twitter: Integration of Diverse Features." *The Nucleus*.

Conference Presentations

- **Owais, M.**, Nakajima, K., and Aida, M. (2025). "Text-Based Social Bot Detection Leveraging Linguistic Structures and Sentiment Features." In *Proceedings of the 14th International Conference on Complex Networks and their Applications*.
- Shaikh, J. A., Wang, C., **Owais, M.**, Zia, U., Sima, M. W. U., & Arshad, M. (2024, December). "FOID: A Feature-Optimized Intrusion Detection System for Securing IoMT Healthcare Networks." In *18th International Conference on Open Source Systems and Technologies (ICOSST)*, IEEE.

Working Papers and Work in Progress

- **Owais, M.**, Nakajima, K., and Aida, M. (2026). "Detecting Evolving Social Bots Through Hybrid Linguistic and Profile-Based Feature Integration." *PLOS ONE*.
- **Owais, M.**, Sajid, M., and Malik, U. H. (2026). "Self-Jailbreaking Large Language Models: Multi-Turn Adversarial Prompting and Cross-Model Guardrail Failures." Journal under preparation.

Professional Experience

Teaching Experience

- Sep 2024 – Jan 2025 ■ **University of Engineering and Technology (UET), Lahore**
Course Instructor (Undergraduate)
Application to information and communication technology (1st Semester)
- Aug 2023 – Feb 2025 ■ **Superior University Lahore**, Computer Science
Course Designer and Instructor (Undergraduate)
Programming Fundamentals (1st Semester), Object-Oriented Programming (2nd Semester), Introduction to Logic (1st Semester), Data Structures and Algorithms (4th Semester)

Research Experience

- Aug 2022 – Oct 2023 ■ **Center for Language Engineering (CLE), UET Lahore**
Research Officer
- Mar 2022 – Aug 2022 ■ **Data Science Lab (DSL), UET Lahore**
Research Intern

Industry Experience

- Jul 2021 – Mar 2022 ■ **CodeFreaks**, Lahore
Junior React Native Developer
- Jan 2021 – June 2021 ■ **LearnToEarn**, Mianwali
Java Intern

Research Projects

- “*E-Khidmat Voice Assistant Application*”, Center for Language Engineering (CLE), Lahore.
Voice-enabled Urdu query system providing spoken and text responses for public service information. (2023)
- “*Automated Meeting Minutes Generation System*”, Center for Language Engineering (CLE), Lahore.
Urdu-dominant multilingual speech processing system for automatic extraction and generation of meeting minutes from mixed Urdu–English discussions. (2022 – 2023)
- “*Urdu Speech Recognition System*”, Center for Language Engineering (CLE), Lahore.
Automatic speech recognition system designed for continuous Urdu speech using broadcast and conversational data. (2022 – 2023)
- “*Speaker Identification from Broadcast Speech Dataset*”, Center for Language Engineering (CLE), Lahore.
Speaker recognition system developed using broadcast data of 207 renowned Pakistani personalities across multiple domains. (2022 – 2023)

Presentations, Talks, & Academic Services

- 2026 ■ Presented a research poster titled “**Decoding Non-Human Language: Detecting Bots and AI-Generated Text on Reddit**” at the **System Design Poster Presentation**, Tokyo Metropolitan University (TMU).
- 2025 ■ Participated as a judge for project evaluations at the **Information Technology Exhibition & Competition (ITEC)**, UET, an in-house talent hunt and technical event organized in collaboration with industrial partners.
- 2024 ■ Delivered a session at **Technical Education & Vocational Training Authority (TEVTA)** on Machine Learning & AI: Concepts and Applications, focusing on practical perspectives and real-world applications.

Presentations, Talks, & Academic Services (continued)

- Delivered a session at the **Technical Education & Vocational Training Authority (TEVTA)**, Lahore, on the importance of Python and its applications.
- Accompanied students from Superior University Lahore to CodeEx 2024, a Speed Programming Competition hosted at **Namal University, Mianwali**.

Mentorship

- *Hamad & Sana*, Department of Computer Science, Superior University Lahore (2023–2024). Supervised undergraduate research project titled “Advancing Urdu Speech Recognition through Dataset Construction and Fine-Tuning.”
- *Ahmed & Mujtaba*, Department of Computer Science, Superior University Lahore (2023–2024). Supervised undergraduate research project titled “Leveraging Phonetic Algorithms and Word Embeddings for Improved Roman Urdu Sentiment Analysis.”
- *Umair & Sidra*, Department of Computer Science, Superior University Lahore (2024). Supervised undergraduate research project titled “Action Item Identification Using Multimodal Meeting Transcripts.”

Awards and Achievements

- 2025 ■ **JSPS Research Funding.**
Awarded for doctoral research support in Japan.
- **Tokyo Global Partner (TGP) Scholarship.**
Fully funded scholarship awarded for Ph.D. studies at Tokyo Metropolitan University, Japan.
- 2023 ■ **Best Poster Presentation Award.**
Awarded by the Department of Computer Science, University of Engineering and Technology (UET), Lahore.
- 2022 ■ **Prime Minister’s Youth Laptop Scheme.**
Awarded based on high academic merit in the department.
- 2021 ■ **Merit-Based Scholarship.**
Awarded during M.S. studies for outstanding academic performance.
- 2017 ■ **Fully Funded Undergraduate Scholarship.**
Awarded for four-year Bachelor’s degree based on academic excellence.

Certification & Courses

- 2026 ■ **Transformers & Large Language Models**
Stanford University
- 2025 ■ **Gen AI Foundational Models for NLP & Language Understanding**
IBM
- **Deep Learning and Reinforcement Learning**
IBM
- **NLP: Twitter Sentiment Analysis**
Coursera
- 2024 ■ **Machine Learning**
Stanford University

Certification & Courses (continued)

2022 ■ **Python Core**
Sololearn

Skills

Coding ■ Python / Django / AI / ML / DL / NLP / PHP / WordPress / HTML / CSS / C / C++ / Redux / ECMAScript / SQL / React Native

Tools ■ Git / Microsoft Office / Kaldi Toolkit / Firebase / Shell (Bash) / $\text{\LaTeX}$ (Overleaf) / R Markdown / Ubuntu

Misc. ■ Academic Research / Teaching / Training / Consultation

Research Interests

Computational and data-driven approaches for Social Network Analysis and Intelligent Systems, Natural Language Processing (NLP), Large Language Models (LLMs), LLM Security, Speech Recognition, Speaker Identification, and Multimodal Language Understanding. My research further explores Deep Learning (DL) and Machine Learning (ML) methodologies.

Languages

English ■ Business

Urdu ■ Native

Saraiki ■ Native

Hindi ■ Conversational